Many powerful numbers between consecutive powers

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Abstract

Let $\kappa \geq 2$ and let $h_\kappa(n)$ denote the number of κ-full integers in the open interval $(n^\kappa, (n+1)^\kappa)$. De Koninck and Luca, and De Koninck–Luca–Shparlinski, proved that $h_\kappa(n) \geq (\log n)^{1/3+o(1)}$ for infinitely many n. We show by a refinement of their method — replacing a self-referential approximation condition by a fixed-tolerance application of Dirichlet's theorem, and choosing the auxiliary squarefree numbers as the divisors of a primorial — that for every fixed $\kappa \geq 2$,

$$h_\kappa(n) \ \gg_\kappa \ \frac{\log n}{\log\log n \ \log\log\log n}$$

for infinitely many n. In particular, in Erdős's question whether $h_2(n) \asymp (\log n)^{c \pm o(1)}$, any admissible exponent satisfies $c \geq 1$.

1 Introduction

An integer $m \geq 1$ is *κ-full* (for $\kappa = 2$, *powerful* or *squarefull*) if $p^\kappa \mid m$ for every prime $p \mid m$. Since the number of powerful integers up to x is $\sim \frac{\zeta(3/2)}{\zeta(3)}\sqrt{x}$ [1], the interval $(n^2, (n+1)^2)$ contains on average about $\frac{\zeta(3/2)}{\zeta(3)} - 1 \approx 1.17$ powerful numbers besides none being a perfect square. Erdős asked (see [2, Problem #942]) for the maximal order of

$$h_\kappa(n) := \#\{m \in (n^\kappa, (n+1)^\kappa) : \ m \text{ is } \kappa\text{-full}\},$$

and in particular whether there is a constant $c > 0$ with $h_2(n) < (\log n)^{c+o(1)}$ for all n and $h_2(n) > (\log n)^{c-o(1)}$ infinitely often.

De Koninck and Luca [3] proved $h_2(n) > \frac{9}{20}(\log n/\log\log n)^{1/3}$ for infinitely many n; De Koninck, Luca and Shparlinski [4] extended this to every fixed $\kappa \geq 2$ (indeed uniformly in $\kappa = \kappa(N)$ growing slowly); we are not aware of a previous improvement of these lower bounds. We prove:

Theorem 1.1. *For every fixed $\kappa \geq 2$ there is $c_\kappa > 0$ such that*

$$h_\kappa(n) \ \geq \ c_\kappa \frac{\log n}{\log\log n \ \log\log\log n}$$

for infinitely many positive integers n.

Thus any exponent c as in Erdős's question satisfies $c \ge 1$. The proof refines the method of [3, 4] in two ways. First, those papers impose on the common denominator q a self-referential approximation condition ($|\alpha_j - r_j/q| \le q^{-1-1/2\ell}$ for all j, with q minimal), whose analysis requires Roth's theorem and yields $\log n \asymp \ell^3 \log \ell$ for ℓ constructed κ-full numbers; but the placement of the constructed numbers in the target interval only requires the *fixed* tolerance $\|q\alpha_j\| \le c/(Dd_j^{1/\kappa})$, which Dirichlet's theorem supplies directly with $\log q \ll \ell \log(D^2)$, and Roth's theorem is not needed. Second, the auxiliary squarefree numbers d_j need not be the first 2ℓ squarefree integers with $D = \prod_j d_j$: it suffices that each d_j divides D, so choosing D to be a primorial and the d_j its squarefree divisors reduces $\log D$ from $\Theta(\ell \log \ell)$ to $O(\log \ell \log\log \ell)$, and the available upper bound for $\log n$ to $O_\kappa(\ell \log \ell \log\log \ell)$.

2 Proof of Theorem 1.1

Fix $\kappa \ge 2$ and a large integer ℓ. Let h be minimal with $2^h - 1 \ge 2\ell$, so $h \asymp \log \ell$, and let

$$D = p_1 p_2 \cdots p_h$$

be the product of the first h primes; by the prime number theorem $\log D \asymp h \log h \asymp \log \ell \log\log \ell$. Let $d_1 < \cdots < d_{2\ell}$ be any 2ℓ of the $2^h - 1$ squarefree divisors $d > 1$ of D, and set $\alpha_j = d_j^{-1/\kappa}$, an irrational number, for $j = 1, \dots, 2\ell$.

Choice of q

By the box principle (Dirichlet's theorem on simultaneous approximation, with the per-coordinate tolerances $\delta_j := \dfrac{1}{2^{\kappa+1}\kappa\, Dd_j^{1/\kappa}}$): subdividing $[0,1)^{2\ell}$ into $\prod_j \lceil \delta_j^{-1} \rceil$ boxes and considering the points $(\{t\alpha_1\}, \dots, \{t\alpha_{2\ell}\})$ for $0 \le t \le \prod_j \lceil \delta_j^{-1} \rceil$, there exist $t_1 > t_2$ in the same box; $q := t_1 - t_2$ satisfies

$$1 \le q \le \prod_{j=1}^{2\ell} \lceil \delta_j^{-1} \rceil \le \left(4\kappa 2^\kappa D^{1+1/\kappa}\right)^{2\ell}, \qquad \|q\alpha_j\| \le \delta_j \quad (j = 1, \dots, 2\ell). \tag{1}$$

In particular $\log q \ll_\kappa \ell \log D \ll \ell \log \ell \, \log\log \ell$.

Let r_j be the nearest integer to $q\alpha_j$ and $\varepsilon_j = q\alpha_j - r_j$, so $|\varepsilon_j| \le \delta_j$ and $\varepsilon_j \ne 0$ (each $\alpha_j = d_j^{-1/\kappa}$ is irrational, as $d_j > 1$ is squarefree). Note $r_j \ge 1$: if $r_j = 0$ then $\|q\alpha_j\| = q\alpha_j \ge \alpha_j = d_j^{-1/\kappa} > \delta_j$, a contradiction.

The constructed numbers

Set $n = Dq$ and

$$m_j = d_j\, D^\kappa r_j^\kappa \qquad (j = 1, \dots, 2\ell).$$

Each m_j is κ-full: every prime of d_j divides D, hence appears in m_j with exponent at least $\kappa + 1$, while primes of D or of r_j appear with exponent at least κ. The m_j are pairwise distinct: comparing p-adic valuations modulo κ in $d_i r_i^\kappa = d_j r_j^\kappa$ forces $v_p(d_i) = v_p(d_j)$ for all p, i.e. $d_i = d_j$.

Placement

Since $d_j\alpha_j^\kappa = 1$,

$$d_j r_j^\kappa = d_j(q\alpha_j - \varepsilon_j)^\kappa = q^\kappa - \kappa q^{\kappa-1} d_j^{1/\kappa}\varepsilon_j + E_j, \qquad |E_j| \le 2^\kappa q^{\kappa-2} d_j^{2/\kappa}\varepsilon_j^2\,\kappa^2,$$

say, for q large. Hence

$$m_j - n^\kappa = D^\kappa\big(d_j r_j^\kappa - q^\kappa\big) = -\kappa D^\kappa q^{\kappa-1} d_j^{1/\kappa}\varepsilon_j + D^\kappa E_j.$$

Write $M_j := \kappa D^\kappa q^{\kappa-1} d_j^{1/\kappa}|\varepsilon_j|$ for the absolute value of the main term. With $|\varepsilon_j| \le \delta_j = \big(2^{\kappa+1}\kappa D d_j^{1/\kappa}\big)^{-1}$ we get $M_j \le 2^{-(\kappa+1)} D^{\kappa-1} q^{\kappa-1} = 2^{-(\kappa+1)} n^{\kappa-1}$, while the error is dominated termwise:

$$\frac{D^\kappa |E_j|}{M_j} \le \frac{2^\kappa \kappa^2\, q^{\kappa-2} d_j^{2/\kappa}\varepsilon_j^2}{\kappa\, q^{\kappa-1} d_j^{1/\kappa}|\varepsilon_j|} = \frac{2^\kappa \kappa\, d_j^{1/\kappa}|\varepsilon_j|}{q} \le \frac{1}{2Dq} \le \frac{1}{2}.$$

Hence $\frac{1}{2}M_j \le |m_j - n^\kappa| \le \frac{3}{2}M_j$; since $\varepsilon_j \ne 0$ this gives

$$0 < |m_j - n^\kappa| \le \tfrac{3}{2}\cdot 2^{-(\kappa+1)} n^{\kappa-1} < n^{\kappa-1} \le \min\big((n+1)^\kappa - n^\kappa,\ n^\kappa - (n-1)^\kappa\big),$$

and the sign of $m_j - n^\kappa$ equals that of the main term, i.e. is opposite to that of ε_j. Therefore each m_j lies in $(n^\kappa, (n+1)^\kappa)$ if $\varepsilon_j < 0$, and in $((n-1)^\kappa, n^\kappa)$ if $\varepsilon_j > 0$. By the pigeonhole principle, at least ℓ of the 2ℓ numbers m_j lie in a single one of these two intervals; setting $N = n$ in the first case and $N = n-1$ in the second,

$$h_\kappa(N) \ \ge\ \ell.$$

Conclusion

Let $X = \log N$ and $f(x) = x/(\log x\, \log\log x)$; one checks f is increasing for $x \ge 16$. We have shown $X \le Y_\ell := C_\kappa\, \ell \log\ell\, \log\log\ell$ with $C_\kappa \ge 1$, and $X \to \infty$ as $\ell \to \infty$ (note $N \ge D-1$). Since $Y_\ell \ge \ell$, we have $\log Y_\ell \ge \log\ell$ and $\log\log Y_\ell \ge \log\log\ell$, so monotonicity gives, for ℓ large,

$$f(X) \ \le\ f(Y_\ell) = \frac{C_\kappa\, \ell\, \log\ell\, \log\log\ell}{\log Y_\ell\, \log\log Y_\ell} \ \le\ C_\kappa\, \ell, \qquad \text{i.e.} \qquad \ell \ \gg_\kappa\ \frac{\log N}{\log\log N\, \log\log\log N}.$$

Letting $\ell \to \infty$ produces infinitely many such N. □

Remark 2.1. The construction is completely explicit and was verified computationally in small cases. For $\kappa = 2$, $h = 2$, using all three nontrivial divisors $2, 3, 6$ of $D = 6$: the smallest admissible q is 485, giving $N = 2909$, and indeed two powerful numbers lie in $(2909^2, 2910^2)$. For $h = 3$ ($D = 30$, all seven divisors $2, 3, 5, 6, 10, 15, 30$): a lattice-reduction search finds q with about 26 decimal digits for which five of the seven constructed numbers lie in $(n^2, (n+1)^2)$ (and two in the lower interval). For $\kappa = 2$, Theorem 1.1 has been formalized and machine-checked in Lean 4/Mathlib, with no `sorry` and only the standard axioms. The development certifies the simultaneous box principle, the κ-fullness and distinctness of the constructed numbers $d_j D^2 r_j^2$ (including the instance $8467200 = 3\cdot 6^2\cdot 280^2$ and $8468064 = 6\cdot 6^2\cdot 198^2$, both powerful and both in $(2909^2, 2910^2)$), the window placement, the count $2^{\omega(D)} - 1$ of squarefree divisors of the primorial, and the size bound $\log\prod_{i<h} p_i \ll h\log h$ — which uses the Chebyshev lower bound $\pi(x) \gg x/\log x$ to bound the hth prime by $O(h\log h)$ — together with the pigeonhole count and the final inversion, yielding $c\,\log n/(\log\log n\, \log\log\log n) \le h_2(n)$ for infinitely many n. The certificate is available at https://github.com/scottdhughes/erdos942.

Remark 2.2. The bottleneck is the box principle: tolerance $e^{-\Theta(\log \ell \log\log \ell)}$ in dimension 2ℓ costs $\log q \asymp \ell \log \ell \log\log \ell$, so Theorem 1.1 may be close to the natural limit of this Dirichlet/primorial construction; removing the remaining $\log\log N \log\log\log N$ factor appears to require either simultaneous approximation of quality beyond Dirichlet's for the specific algebraic vector $(d_j^{-1/\kappa})_j$, or a different construction.

References

[1] P. Bateman and E. Grosswald, On a theorem of Erdős and Szekeres, *Illinois J. Math.* **2** (1958), 88–98.

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[4] J.-M. De Koninck, F. Luca and I. E. Shparlinski, Powerful numbers in short intervals, *Bull. Austral. Math. Soc.* **71** (2005), 11–16.

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